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**HK IMO Training 2023-2024**  
**Problem Set 3**  
**For 14 October 2023**

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**Problem 9.** Find all triples  $(p, q, r)$  of prime numbers such that  $p^q + p^r$  is a perfect square.

**Problem 10.** In acute  $\triangle ABC$ , the tangents at  $B$  and  $C$  to the circumcircle  $\Omega$  of  $\triangle ABC$  meet at  $D$ . A line passing through  $D$  cuts the circumcircles of  $\triangle ABD$  and  $\triangle ACD$  again at  $E$  and  $F$  respectively. Prove that the lines  $CE$  and  $BF$  intersect on  $\Omega$ .

**Problem 11.** We first choose a positive integer  $n$ , then write down the polynomial  $(x + 1)^n$  in its expanded form, and change every coefficient to its reciprocal. For example, if we choose  $n = 3$ , then the resultant polynomial is  $x^3 + \frac{1}{3}x^2 + \frac{1}{3}x + 1$  since  $(x + 1)^3 = x^3 + 3x^2 + 3x + 1$ . Prove that we can choose a positive integer  $n$  such that the sum of all coefficients of the resultant polynomial is less than 2.023.

**Problem 12.** There are  $x$  countries and  $y$  cities on a planet, where each country contains at least three cities (and every city belongs to a country). Some of the cities are connected by roads. For each city, it is connected to at least half of the cities in this country, and is connected to exactly one city in another country. For any two countries, there are at most two roads between cities of the two countries. Also, for any two countries with fewer than  $2x$  cities in total, there is at least one road between cities of the two countries. Prove that there exists a cycle of at least  $x + \frac{y}{2}$  cities.